六 二重积分

一、选择题

1.(D). 当$(x,y)\in D$时，$\ln(x+y)\leqslant 0$，$I=\iint\limits_D \ln(x+y)\,dx\,dy<0$；

因为区域D关于$y=x$对称，所以$\iint\limits_D \sin x\cos y\,dx\,dy=\iint\limits_D \sin y\cos x\,dx\,dy$，于是

$$J=2\iint\limits_D \sin x\cos y\,dx\,dy=\iint\limits_D \sin x\cos y\,dx\,dy+\iint\limits_D \sin y\cos x\,dx\,dy=\iint\limits_D \sin(x+y)\,dx\,dy.$$

当$(x,y)\in D$时，$e^{x+y}-1>x+y\geqslant \sin(x+y)\geqslant 0$，则$K>J>0$. 选(D).

2.(D). 积分所对应的直角坐标平面的区域为$D:0\leqslant x\leqslant 1,0\leqslant y\leqslant\sqrt{x-x^2}$. 选(D).

3.(D). 将区域D向y轴投影，划分成区域

$$D_1=\{(x,y)\mid 0\leqslant x\leqslant\sqrt{y},0\leqslant y\leqslant 1\},$$

$$D_2=\{(x,y)\mid 0\leqslant x\leqslant\sqrt{2y-y^2},1\leqslant y\leqslant 2\},$$

则原式$=\int_0^1 dy\int_0^{\sqrt{y}} f(x,y)\,dx+\int_1^2 dy\int_0^{\sqrt{2y-y^2}} f(x,y)\,dx$. 选(D).

4.(B). 令$\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}(0\leqslant\theta\leqslant 2\pi,0\leqslant r\leqslant a)$.

由$\iint\limits_D \sqrt{a^2-x^2-y^2}\,dx\,dy=\int_0^{2\pi}d\theta\int_0^a r\sqrt{a^2-r^2}\,dr=-\pi\int_0^a (a^2-r^2)^{\frac{1}{2}}\,d(a^2-r^2)$

$$=-\pi\cdot\frac{2}{3}(a^2-r^2)^{\frac{3}{2}}\Big|_0^a=\frac{2\pi a^3}{3}=\frac{16\pi}{3},$$

解得$a=2$. 选(B).

5.(B). 由对称性得$I=\iint\limits_D \frac{\sin x\cos y+y}{\sqrt{x^2+y^2}}dx\,dy=\iint\limits_D \frac{y}{\sqrt{x^2+y^2}}dx\,dy$.

令$\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(\frac{\pi}{4}\leqslant\theta\leqslant\frac{3\pi}{4},\csc\theta\leqslant r\leqslant\sqrt{2}\right)$，则

$$I=\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\sin\theta\,d\theta\int_{\csc\theta}^{\sqrt{2}} r\,dr=\frac{1}{2}\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\sin\theta(2-\csc^2\theta)\,d\theta=\frac{1}{2}\left(2\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\sin\theta\,d\theta-\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\csc\theta\,d\theta\right)$$

$$=-\cos\theta\Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}}-\frac{1}{2}\ln|\csc\theta-\cot\theta|\Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}}=\sqrt{2}-\ln(\sqrt{2}+1).$$

选(B).

6.(B). 根据对称性，令$D_1=\{(x,y)\mid 0\leqslant x\leqslant\pi,0\leqslant y\leqslant x\}$，

$$\iint\limits_D \sin x\sin y\cdot\max\{x,y\}\,d\sigma=2\iint\limits_{D_1} x\sin x\sin y\,d\sigma=2\int_0^\pi x\sin x\,dx\int_0^x \sin y\,dy$$

$$=2\int_0^\pi x\sin x(1-\cos x)\,dx=2\int_0^\pi x\sin x\,dx-2\int_0^\pi x\sin x\cos x\,dx$$

$$=\pi\int_0^\pi \sin x\,dx-\int_0^\pi x\,d(\sin^2 x)=\frac{5\pi}{2}.$$

选(B).

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二、填空题

1. 解 $\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}\sum\limits_{j=1}^{2n}\dfrac{1}{n^2+i^2}\cdot\left(\dfrac{j}{n}+\sin\dfrac{\pi j}{n}\right)\left(1+\cos\dfrac{\pi j}{n}\right)^2$

$=\lim\limits_{n\to\infty}\dfrac{1}{n^2}\sum\limits_{i=1}^{n}\sum\limits_{j=1}^{2n}\dfrac{1}{1+\left(\dfrac{i}{n}\right)^2}\cdot\left(\dfrac{j}{n}+\sin\dfrac{\pi j}{n}\right)\left(1+\cos\dfrac{\pi j}{n}\right)^2$

$=\int_0^1\dfrac{\mathrm{d}x}{1+x^2}\int_0^2(y+\sin\pi y)(1+\cos\pi y)^2\mathrm{d}y$

$=\dfrac{1}{4}\int_0^2\left(\dfrac{1}{\pi}\cdot\pi y+\sin\pi y\right)(1+\cos\pi y)^2\mathrm{d}(\pi y)=\dfrac{1}{4}\int_0^{2\pi}\left(\dfrac{t}{\pi}+\sin t\right)(1+\cos t)^2\mathrm{d}t$

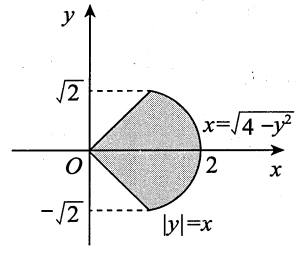
$\xlongequal{t-\pi=u}\dfrac{1}{4}\int_{-\pi}^{\pi}\left(\dfrac{\pi+u}{\pi}-\sin u\right)(1-\cos u)^2\mathrm{d}u=\dfrac{1}{2}\int_0^{\pi}(1-\cos u)^2\mathrm{d}u$

$=\int_0^{\pi}\left(2\sin^2\dfrac{u}{2}\right)^2\mathrm{d}\left(\dfrac{u}{2}\right)=4\int_0^{\frac{\pi}{2}}\sin^4u\,\mathrm{d}u=4\cdot\dfrac{3}{4}\cdot\dfrac{1}{2}\cdot\dfrac{\pi}{2}=\dfrac{3\pi}{4}.$

2. 解 $\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}\sum\limits_{j=1}^{2n}\dfrac{i+j}{n^3}f\left(\dfrac{2i-j}{n}\right)=\lim\limits_{n\to\infty}\dfrac{1}{n^2}\sum\limits_{i=1}^{n}\sum\limits_{j=1}^{2n}\left(\dfrac{i}{n}+\dfrac{j}{n}\right)f\left(2\dfrac{i}{n}-\dfrac{j}{n}\right)$

$=\int_0^1\mathrm{d}x\int_0^2(x+y)f(2x-y)\mathrm{d}y.$

3. 解 积分区域如图所示，显然关于 x 轴对称，由二重积分的奇偶性得 $\int_{-\sqrt{2}}^{\sqrt{2}}\mathrm{d}y\int_{|y|}^{\sqrt{4-y^2}}\dfrac{x+\sin^3xy}{4+x^2+y^2}\mathrm{d}x=\int_{-\sqrt{2}}^{\sqrt{2}}\mathrm{d}y\int_{|y|}^{\sqrt{4-y^2}}\dfrac{x}{4+x^2+y^2}\mathrm{d}x$，



第 3 题图

令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(-\dfrac{\pi}{4}\leqslant\theta\leqslant\dfrac{\pi}{4},0\leqslant r\leqslant2\right)$，则

$\int_{-\sqrt{2}}^{\sqrt{2}}\mathrm{d}y\int_{|y|}^{\sqrt{4-y^2}}\dfrac{x}{4+x^2+y^2}\mathrm{d}x=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\cos\theta\,\mathrm{d}\theta\int_0^2\dfrac{r^2}{4+r^2}\mathrm{d}r$

$=\sqrt{2}\int_0^2\left(1-\dfrac{4}{4+r^2}\right)\mathrm{d}r$

$=\sqrt{2}\left(2-2\arctan\dfrac{r}{2}\Big|_0^2\right)=\sqrt{2}\left(2-\dfrac{\pi}{2}\right).$

4. 解 改变积分次序得 $\int_0^2\mathrm{d}x\int_x^2\mathrm{e}^{\frac{x}{y}}\mathrm{d}y=\int_0^2\mathrm{d}y\int_0^y\mathrm{e}^{\frac{x}{y}}\mathrm{d}x=\int_0^2(\mathrm{e}-1)y\,\mathrm{d}y=2\mathrm{e}-2.$

5. 解 令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(0\leqslant\theta\leqslant\dfrac{\pi}{4},0\leqslant r\leqslant2\sqrt{\cos2\theta}\right)$，则

$\iint\limits_D2xy\,\mathrm{d}x\,\mathrm{d}y=\int_0^{\frac{\pi}{4}}2\sin\theta\cos\theta\,\mathrm{d}\theta\int_0^{2\sqrt{\cos2\theta}}r^3\mathrm{d}r=4\int_0^{\frac{\pi}{4}}\sin2\theta\cos^22\theta\,\mathrm{d}\theta$

$=2\int_0^{\frac{\pi}{4}}\sin2\theta\cos^22\theta\,\mathrm{d}(2\theta)=2\int_0^{\frac{\pi}{2}}\sin\theta\cos^2\theta\,\mathrm{d}\theta$

$=2\int_0^{\frac{\pi}{2}}\sin\theta(1-\sin^2\theta)\mathrm{d}\theta=2\left(1-\dfrac{2}{3}\right)=\dfrac{2}{3}.$

6. 解 由对称性得

$\iint\limits_D(x^2+2y)\mathrm{d}x\,\mathrm{d}y=\iint\limits_{x^2+(2y)^2\leqslant1}x^2\mathrm{d}x\,\mathrm{d}y=\dfrac{1}{2}\iint\limits_{x^2+(2y)^2\leqslant1}x^2\mathrm{d}x\,\mathrm{d}(2y)=\dfrac{1}{2}\iint\limits_{x^2+y^2\leqslant1}x^2\mathrm{d}x\,\mathrm{d}y$

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$$=\frac{1}{4}\iint\limits_{x^2+y^2\leqslant 1}(x^2+y^2)\mathrm{d}x\,\mathrm{d}y=\frac{\pi}{2}\int_0^1 r^3\mathrm{d}r=\frac{\pi}{8}.$$

7. 解 令$\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(0\leqslant\theta\leqslant\frac{\pi}{4},1\leqslant r\leqslant\sec\theta\right)$,则

$$\iint\limits_D\frac{x}{\sqrt{x^2+y^2}}\mathrm{d}x\,\mathrm{d}y=\int_0^{\frac{\pi}{4}}\cos\theta\mathrm{d}\theta\int_1^{\sec\theta}r\mathrm{d}r=\frac{1}{2}\int_0^{\frac{\pi}{4}}\cos\theta(\sec^2\theta-1)\mathrm{d}\theta=\frac{1}{2}\int_0^{\frac{\pi}{4}}(\sec\theta-\cos\theta)\mathrm{d}\theta$$

$$=\frac{1}{2}\ln|\sec\theta+\tan\theta|\Big|_0^{\frac{\pi}{4}}-\frac{\sqrt{2}}{4}=\frac{1}{2}\ln(\sqrt{2}+1)-\frac{\sqrt{2}}{4}.$$

8. 解 令$\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(\frac{\pi}{4}\leqslant\theta\leqslant\pi,0\leqslant r\leqslant 2\sin\theta\right)$,则

$$I=\iint\limits_D\frac{\mathrm{d}\sigma}{\sqrt{x^2+y^2}\cdot\sqrt{4-x^2-y^2}}=\int_{\frac{\pi}{4}}^{\pi}\mathrm{d}\theta\int_0^{2\sin\theta}\frac{\mathrm{d}r}{\sqrt{4-r^2}},$$

由$\int_0^{2\sin\theta}\frac{\mathrm{d}r}{\sqrt{4-r^2}}\xlongequal{r=2\sin t}\int_0^{\theta}\frac{2\cos t\,\mathrm{d}t}{2\cos t}=\theta$,得$I=\int_{\frac{\pi}{4}}^{\pi}\theta\mathrm{d}\theta=\frac{15\pi^2}{32}$.

9. 解 由积分区域关于$y=x$对称,得$I=\iint\limits_D\frac{f(x)}{f(x)+f(y)}\mathrm{d}x\,\mathrm{d}y=\iint\limits_D\frac{f(y)}{f(y)+f(x)}\mathrm{d}x\,\mathrm{d}y$,则

$$I=\frac{1}{2}\left[\iint\limits_D\frac{f(x)}{f(x)+f(y)}\mathrm{d}x\,\mathrm{d}y+\iint\limits_D\frac{f(y)}{f(y)+f(x)}\mathrm{d}x\,\mathrm{d}y\right]=\frac{1}{2}\iint\limits_D\mathrm{d}x\,\mathrm{d}y=\frac{1}{2}\cdot\frac{1}{4}(4\pi-\pi)=\frac{3\pi}{8}.$$

10. 解 由$f(x)g(y-x)=\begin{cases}a^2, & 0\leqslant x\leqslant 1,0\leqslant y-x\leqslant 1,\\ 0, & \text{其他},\end{cases}$得

$$I=\iint\limits_D f(x)g(y-x)\mathrm{d}x\,\mathrm{d}y=a^2\int_0^1\mathrm{d}x\int_x^{x+1}\mathrm{d}y=a^2.$$

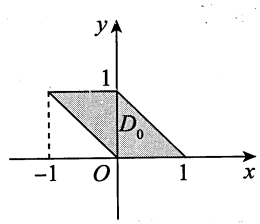
11. 解 在$D_1=\{(x,y)\mid-\infty<x<+\infty,0\leqslant y\leqslant 1\}$上,$f(y)=y$;

在$D_2=\{(x,y)\mid 0\leqslant x+y\leqslant 1\}$上,$f(x+y)=x+y$,则

$D_0=D_1\cap D_2=\{(x,y)\mid -y\leqslant x\leqslant 1-y,0\leqslant y\leqslant 1\}$,

如图所示,$f(y)f(x+y)=y(x+y)$,所以

$$\iint\limits_D f(y)f(x+y)\mathrm{d}x\,\mathrm{d}y=\int_0^1\mathrm{d}y\int_{-y}^{1-y}y(x+y)\mathrm{d}x=\frac{1}{4}.$$



第 11 题图

12. 解 当$a\to 0^+$时,

$$\mathrm{e}^a-\mathrm{e}^{\ln(1+a)}=\mathrm{e}^{\ln(1+a)}\cdot[\mathrm{e}^{a-\ln(1+a)}-1]\sim a-\ln(1+a)\sim\frac{1}{2}a^2.$$

由$\lim\limits_{\substack{x\to 1\\ y\to -1}}\frac{f(x,y)-2x-y-3}{(x-1)^2+(y+1)^2}=1$,得$f(1,-1)=4$.

再由二重积分中值定理,得$\iint\limits_D f(x,y)\mathrm{d}\sigma=f(\xi,\eta)\cdot\pi\cdot a\cdot\frac{a}{2}=\frac{\pi a^2}{2}f(\xi,\eta),f(\xi,\eta)\in D$,则

$$\lim_{a\to 0^+}\frac{\iint\limits_D f(x,y)\mathrm{d}\sigma}{\mathrm{e}^a-\mathrm{e}^{\ln(1+a)}}=\lim_{a\to 0^+}\frac{\frac{\pi a^2}{2}}{\frac{1}{2}a^2}f(\xi,\eta)=\pi f(1,-1)=4\pi.$$

13. 解 由$\lim\limits_{\substack{x\to 1\\ y\to 0}}\frac{f(x,y)-2x-2}{\sin\pi x+y^2}=-1$,得$\lim\limits_{\substack{x\to 1\\ y\to 0}}[f(x,y)-2x-2]=0$,即$\lim\limits_{\substack{x\to 1\\ y\to 0}}f(x,y)=4$.

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因为 $f(x,y)$ 连续，所以 $f(1,0)=4$.

令 $D=\{(x,y)\mid (x-1)^2+4y^2\leqslant r^2\}=\left\{(x,y)\left|\frac{(x-1)^2}{r^2}+\frac{y^2}{\left(\frac{r}{2}\right)^2}\leqslant 1\right.\right\}$.

由积分中值定理，存在 $(\xi,\eta)\in D$，使得 $\iint\limits_{(x-1)^2+4y^2\leqslant r^2} f(x,y)\mathrm{d}x\mathrm{d}y=f(\xi,\eta)\cdot\pi\cdot r\cdot\frac{r}{2}$.

又当 $r\to 0$ 时，$\int_0^{e^{r^2}-1}\frac{\tan 2t}{\arcsin t}\mathrm{d}t\sim\int_0^{r^2}2\mathrm{d}t=2r^2$，则

$$\lim_{r\to 0}\frac{\iint\limits_{(x-1)^2+4y^2\leqslant r^2} f(x,y)\mathrm{d}x\mathrm{d}y}{\int_0^{e^{r^2}-1}\frac{\tan 2t}{\arcsin t}\mathrm{d}t}=\lim_{r\to 0}\frac{\frac{\pi r^2}{2}\cdot f(\xi,\eta)}{2r^2}=\frac{\pi}{4}\lim_{r\to 0}f(\xi,\eta)=\frac{\pi}{4}f(1,0)=\pi.$$

14. 解 $\int_0^r tf(r^2-t^2)\mathrm{d}t=-\frac{1}{2}\int_0^r f(r^2-t^2)\mathrm{d}(r^2-t^2)=\frac{1}{2}\int_0^{r^2}f(u)\mathrm{d}u$；

令 $D=\{(x,y)\mid x^2+y^2\leqslant r^2\}$，由积分中值定理得，存在 $(\xi,\eta)\in D$，使得

$$\iint\limits_{x^2+y^2\leqslant r^2}\cos(x+y)\mathrm{d}\sigma=\pi r^2\cos(\xi+\eta);$$

$$\text{原式}=\frac{1}{2\pi}\lim_{r\to 0}\frac{\int_0^{r^2}f(u)\mathrm{d}u}{r^2}\cdot\frac{1}{\cos(\xi+\eta)}=\frac{1}{2\pi}\lim_{r\to 0}\frac{\int_0^{r^2}f(u)\mathrm{d}u}{r^2}=\frac{1}{2\pi}\lim_{r\to 0}\frac{2rf(r^2)}{2r}=\frac{f(0)}{2\pi}.$$

15. 解 $\int_0^{\pi}f(x)\mathrm{d}x=\int_0^{\pi}\mathrm{d}x\int_0^x\frac{\sin t}{\pi-t}\mathrm{d}t=\int_0^{\pi}\mathrm{d}t\int_t^{\pi}\frac{\sin t}{\pi-t}\mathrm{d}x=\int_0^{\pi}\sin t\,\mathrm{d}t=2.$

16. 解 $\int_u^1 vf(u^2-v^2)\mathrm{d}v=-\frac{1}{2}\int_u^1 f(u^2-v^2)\mathrm{d}(u^2-v^2)=-\frac{1}{2}\int_0^{u^2-1}f(t)\mathrm{d}t$，则

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_0^x\mathrm{d}u\int_u^1 vf(u^2-v^2)\mathrm{d}v=-\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}\int_0^x\mathrm{d}u\int_0^{u^2-1}f(t)\mathrm{d}t=-\frac{1}{2}\int_0^{x^2-1}f(t)\mathrm{d}t,$$

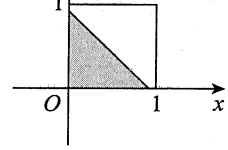
$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\int_0^x\mathrm{d}u\int_u^1 vf(u^2-v^2)\mathrm{d}v=-xf(x^2-1).$$

三、解答题

1. 解 令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(0\leqslant\theta\leqslant\frac{\pi}{4},0\leqslant r\leqslant\frac{\sin\theta}{\cos^2\theta}\right)$，则

$$\int_0^1\mathrm{d}x\int_{x^2}^x(x^2+y^2)^{-\frac{1}{2}}\mathrm{d}y=\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_0^{\frac{\sin\theta}{\cos^2\theta}}\mathrm{d}r=\int_0^{\frac{\pi}{4}}\frac{\sin\theta}{\cos^2\theta}\mathrm{d}\theta=\frac{1}{\cos\theta}\Big|_0^{\frac{\pi}{4}}=\sqrt{2}-1.$$

2. 解 积分区域 D 如图中阴影部分所示，D 的范围随 t 变化，而 t 又随 x 和 y 变化，故需讨论 t 的取值范围.



第 2 题图

当 $t<0$ 时，$F(t)=0$；

当 $0\leqslant t<1$ 时，$F(t)=\iint\limits_D 1\mathrm{d}x\mathrm{d}y=\frac{1}{2}t^2$；

当 $1\leqslant t<2$ 时，$F(t)=\iint\limits_D f(x,y)\mathrm{d}x\mathrm{d}y=1-\frac{1}{2}(2-t)^2$；

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当 $t \geqslant 2$ 时，$F(t)=1$，则 $F(t)=\begin{cases}0, & t<0,\\ \frac{1}{2}t^2, & 0\leqslant t<1,\\ 1-\frac{1}{2}(2-t)^2, & 1\leqslant t<2,\\ 1, & t\geqslant 2.\end{cases}$

3. 解 令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}(0\leqslant\theta\leqslant 2\pi, 0\leqslant r\leqslant t)$，由

$$F(t)=\int_0^{2\pi}\mathrm{d}\theta\int_0^t rf(r^2)\mathrm{d}r=2\pi\int_0^t rf(r^2)\mathrm{d}r\xlongequal{r^2=u}\pi\int_0^{t^2}f(u)\mathrm{d}u,$$

得 $F'(t)=2\pi tf(t^2)$，$F'(0)=0$，

$$F''(0)=\lim_{t\to 0}\frac{F'(t)-F'(0)}{t}=\lim_{t\to 0}\frac{2\pi tf(t^2)}{t}=2\pi f(0)=2\pi.$$

4. 解 $D:x^2+y^2\leqslant 2x+2y-1$ 可化为 $D:(x-1)^2+(y-1)^2\leqslant 1$.

令 $\begin{cases}x=1+r\cos t,\\ y=1+r\sin t\end{cases}(0\leqslant t\leqslant 2\pi, 0\leqslant r\leqslant 1)$，则

$$\iint_D(x+y^2)\mathrm{d}x\,\mathrm{d}y=\int_0^{2\pi}\mathrm{d}t\int_0^1(1+r\cos t+1+2r\sin t+r^2\sin^2 t)r\mathrm{d}r$$

$$=\int_0^{2\pi}\left(1+\frac{1}{4}\sin^2 t\right)\mathrm{d}t=2\pi+\int_0^{\frac{\pi}{2}}\sin^2 t\,\mathrm{d}t=2\pi+\frac{\pi}{4}=\frac{9\pi}{4}.$$

5. 解 $\iint_D(x^2+y^2)\mathrm{d}x\,\mathrm{d}y=\iint_{x^2+y^2\leqslant 4}(x^2+y^2)\mathrm{d}x\,\mathrm{d}y-\iint_{x^2+y^2\leqslant 2x}(x^2+y^2)\mathrm{d}x\,\mathrm{d}y.$

令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}(0\leqslant\theta\leqslant 2\pi, 0\leqslant r\leqslant 2)$，而 $\iint_{x^2+y^2\leqslant 4}(x^2+y^2)\mathrm{d}x\,\mathrm{d}y=\int_0^{2\pi}\mathrm{d}\theta\int_0^2 r^3\mathrm{d}r=8\pi$，

$$\iint_{x^2+y^2\leqslant 2x}(x^2+y^2)\mathrm{d}x\,\mathrm{d}y=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{2\cos\theta}r^3\mathrm{d}r=8\int_0^{\frac{\pi}{2}}\cos^4\theta\,\mathrm{d}\theta=\frac{3\pi}{2},$$

所以 $\iint_D(x^2+y^2)\mathrm{d}x\,\mathrm{d}y=\frac{13\pi}{2}$.

6. 解 令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(0\leqslant\theta\leqslant\frac{\pi}{2}, 0\leqslant r\leqslant 1\right)$，则

$$\iint_D\sqrt{\frac{1-x^2-y^2}{1+x^2+y^2}}\mathrm{d}x\,\mathrm{d}y=\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^1\sqrt{\frac{1-r^2}{1+r^2}}r\mathrm{d}r=\frac{\pi}{4}\int_0^1\sqrt{\frac{1-r^2}{1+r^2}}\mathrm{d}r^2\xlongequal{r^2=t}\frac{\pi}{4}\int_0^1\sqrt{\frac{1-t}{1+t}}\mathrm{d}t,$$

令 $\frac{1-t}{1+t}=u^2$，则 $t=\frac{1-u^2}{1+u^2}$，$\mathrm{d}t=\frac{-4u}{(1+u^2)^2}\mathrm{d}u$，

$$\text{原式}=\frac{\pi}{4}\int_1^0\frac{-4u^2}{(1+u^2)^2}\mathrm{d}u=\pi\int_0^1\frac{u^2}{(1+u^2)^2}\mathrm{d}u=\pi\int_0^1\left[\frac{1}{1+u^2}-\frac{1}{(1+u^2)^2}\right]\mathrm{d}u$$

$$=\pi\int_0^1\frac{1}{1+u^2}\mathrm{d}u-\pi\int_0^1\frac{1}{(1+u^2)^2}\mathrm{d}u=\frac{\pi^2}{4}-\pi\int_0^1\frac{1}{(1+u^2)^2}\mathrm{d}u.$$

因为 $\int_0^1\frac{1}{(1+u^2)^2}\mathrm{d}u\xlongequal{u=\tan\theta}\int_0^{\frac{\pi}{4}}\frac{1}{\sec^4\theta}\cdot\sec^2\theta\,\mathrm{d}\theta=\int_0^{\frac{\pi}{4}}\cos^2\theta\,\mathrm{d}\theta=\frac{1}{2}\int_0^{\frac{\pi}{4}}(1+\cos 2\theta)\mathrm{d}\theta$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}}(1+\cos 2\theta)\mathrm{d}\theta=\frac{\pi}{8}+\frac{1}{4},$$

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所以原式 $=\frac{\pi^2}{4}-\pi\left(\frac{\pi}{8}+\frac{1}{4}\right)=\frac{\pi^2}{8}-\frac{\pi}{4}$.

7. 解 令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta,\end{cases}$ 则 $I=\int_0^{\frac{3\pi}{4}}\mathrm{d}\theta\int_0^2 r^3\,\mathrm{d}r-\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{2\cos\theta}r^3\,\mathrm{d}r=\frac{9\pi}{4}$.

8. 解 $I=\iint\limits_D\frac{\mathrm{d}x\,\mathrm{d}y}{\sqrt{x^2+y^2}\cdot\sqrt{4-x^2-y^2}}$,其中

$$D=\{(x,y)\mid 0\leqslant x\leqslant 1,-x\leqslant y\leqslant\sqrt{1-x^2}-1\},$$

令 $\begin{cases}x=r\cos t,\\ y=r\sin t,\end{cases}$ 则 $D=\left\{(r,t)\,\middle|\,-\frac{\pi}{4}\leqslant t\leqslant 0,0\leqslant r\leqslant -2\sin t\right\}$,于是

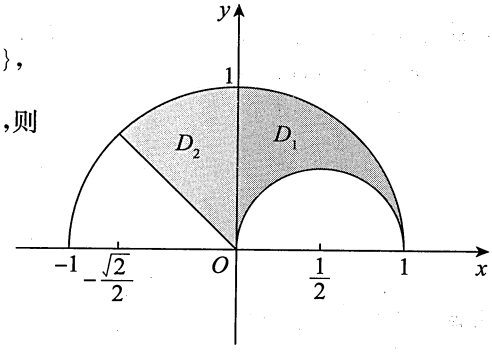
$$I=\iint\limits_D\frac{\mathrm{d}x\,\mathrm{d}y}{\sqrt{x^2+y^2}\cdot\sqrt{4-x^2-y^2}}=\int_{-\frac{\pi}{4}}^0\mathrm{d}t\int_0^{-2\sin t}\frac{\mathrm{d}r}{\sqrt{4-r^2}}=\int_{-\frac{\pi}{4}}^0-t\,\mathrm{d}t=\frac{\pi^2}{32}.$$

9. 解 如图所示,将 D 分成两部分 D_1,D_2,其中

$$D_1=\{(x,y)\mid 0\leqslant x\leqslant 1,\sqrt{x-x^2}\leqslant y\leqslant\sqrt{1-x^2}\},$$

$$D_2=\left\{(x,y)\,\middle|\,-\frac{\sqrt{2}}{2}\leqslant x\leqslant 0,-x\leqslant y\leqslant\sqrt{1-x^2}\right\},\text{则}$$

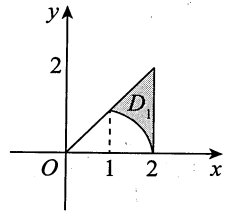
$$\begin{aligned}I&=\int_0^1 x\,\mathrm{d}x\int_{\sqrt{x-x^2}}^{\sqrt{1-x^2}}y\,\mathrm{d}y+\int_{-\frac{\sqrt{2}}{2}}^0 x\,\mathrm{d}x\int_{-x}^{\sqrt{1-x^2}}y\,\mathrm{d}y\\&=\frac{1}{2}\int_0^1 x(1-x)\,\mathrm{d}x+\frac{1}{2}\int_{-\frac{\sqrt{2}}{2}}^0 x(1-2x^2)\,\mathrm{d}x\\&=\frac{1}{48}.\end{aligned}$$



第 9 题图

10. 解 令 $D_1=\{(x,y)\mid 1\leqslant x\leqslant 2,\sqrt{2x-x^2}\leqslant y\leqslant x\}$,如图所示,则

$$\begin{aligned}\iint\limits_D f(x,y)\,\mathrm{d}x\,\mathrm{d}y&=\iint\limits_{D_1}x^2y\,\mathrm{d}x\,\mathrm{d}y=\int_1^2 x^2\,\mathrm{d}x\int_{\sqrt{2x-x^2}}^x y\,\mathrm{d}y\\&=\int_1^2(x^4-x^3)\,\mathrm{d}x=\frac{49}{20}.\end{aligned}$$



第 10 题图

11. 解 根据对称性得 $\iint\limits_D(x^2+4x+y^2)\,\mathrm{d}x\,\mathrm{d}y=4\iint\limits_{D_1}(x^2+y^2)\,\mathrm{d}x\,\mathrm{d}y$,其中

D_1 是 D 位于第一象限的区域.

令 $\begin{cases}x=r\cos\theta,\\ y=r\sin\theta\end{cases}\left(0\leqslant\theta\leqslant\frac{\pi}{4},0\leqslant r\leqslant a\sqrt{\cos 2\theta}\right)$,则

$$\iint\limits_{D_1}(x^2+y^2)\,\mathrm{d}x\,\mathrm{d}y=\int_0^{\frac{\pi}{4}}\mathrm{d}\theta\int_0^{a\sqrt{\cos 2\theta}}r^3\,\mathrm{d}r=\frac{a^4}{8}\int_0^{\frac{\pi}{4}}\cos^2 2\theta\,\mathrm{d}(2\theta)=\frac{\pi a^4}{32},$$

故 $\iint\limits_D(x^2+4x+y^2)\,\mathrm{d}x\,\mathrm{d}y=\frac{\pi a^4}{8}$.

12. 解 令

$$D_1=\{(x,y)\mid -1\leqslant x\leqslant 1,0\leqslant y\leqslant x^2\},$$

$$D_2=\{(x,y)\mid -1\leqslant x\leqslant 1,x^2\leqslant y\leqslant 2\},$$

则 $I=\iint\limits_D\sqrt{|y-x^2|}\,\mathrm{d}x\,\mathrm{d}y=\iint\limits_{D_1}\sqrt{x^2-y}\,\mathrm{d}x\,\mathrm{d}y+\iint\limits_{D_2}\sqrt{y-x^2}\,\mathrm{d}x\,\mathrm{d}y$.

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$$而\iint_{D_1}\sqrt{x^2-y}\,dx\,dy=-\int_{-1}^{1}dx\int_{0}^{x^2}(x^2-y)^{\frac{1}{2}}d(x^2-y)$$

$$=-\frac{2}{3}\int_{-1}^{1}(x^2-y)^{\frac{3}{2}}\Big|_{0}^{x^2}dx=\frac{2}{3}\int_{-1}^{1}|x|^3dx=\frac{4}{3}\int_{0}^{1}x^3dx=\frac{1}{3};$$

$$\iint_{D_2}\sqrt{y-x^2}\,dx\,dy=\int_{-1}^{1}dx\int_{x^2}^{2}(y-x^2)^{\frac{1}{2}}d(y-x^2)=\frac{2}{3}\int_{-1}^{1}(2-x^2)^{\frac{3}{2}}dx=\frac{4}{3}\int_{0}^{1}(2-x^2)^{\frac{3}{2}}dx$$

$$\xlongequal{x=\sqrt{2}\sin t}\frac{4}{3}\int_{0}^{\frac{\pi}{4}}2\sqrt{2}\cos^3t\cdot\sqrt{2}\cos t\,dt=\frac{16}{3}\int_{0}^{\frac{\pi}{4}}\cos^4t\,dt=\frac{4}{3}\int_{0}^{\frac{\pi}{4}}(1+\cos 2t)^2dt$$

$$=\frac{2}{3}\int_{0}^{\frac{\pi}{2}}(1+\cos t)^2dt=\frac{2}{3}\int_{0}^{\frac{\pi}{2}}(1+2\cos t+\cos^2t)dt$$

$$=\frac{2}{3}\left(\frac{\pi}{2}+2+\frac{\pi}{4}\right)=\frac{\pi}{2}+\frac{4}{3}.$$

$$故\iint_{D}\sqrt{|y-x^2|}\,dx\,dy=\frac{\pi}{2}+\frac{5}{3}.$$

13. 解 令$\sqrt{\frac{3}{16}-x^2-y^2}=2(x^2+y^2)$，解得$x^2+y^2=\frac{1}{8}$，则

$$\iint_{x^2+y^2\leqslant\frac{3}{16}}\min\left\{\sqrt{\frac{3}{16}-x^2-y^2},2(x^2+y^2)\right\}dx\,dy$$

$$=\iint_{x^2+y^2\leqslant\frac{1}{8}}2(x^2+y^2)dx\,dy+\iint_{\frac{1}{8}\leqslant x^2+y^2\leqslant\frac{3}{16}}\sqrt{\frac{3}{16}-x^2-y^2}\,dx\,dy$$

$$=2\int_{0}^{2\pi}d\theta\int_{0}^{\frac{1}{2\sqrt{2}}}r^3dr+\int_{0}^{2\pi}d\theta\int_{\frac{1}{2\sqrt{2}}}^{\frac{\sqrt{3}}{4}}r\sqrt{\frac{3}{16}-r^2}\,dr=\frac{\pi}{64}+\frac{\pi}{96}=\frac{5\pi}{192}.$$

14. 解 设$f(x)$的一个原函数为$F(x)$，则

$$\iint_{D}x[\sqrt{1-x^2}+yf(x^2+y^2)]dx\,dy$$

$$=\int_{-1}^{1}x\,dx\int_{x^3}^{1}[\sqrt{1-x^2}+yf(x^2+y^2)]dy$$

$$=\int_{-1}^{1}x\,dx\int_{x^3}^{1}\sqrt{1-x^2}\,dy+\int_{-1}^{1}x\,dx\int_{x^3}^{1}yf(x^2+y^2)dy$$

$$=\int_{-1}^{1}x\sqrt{1-x^2}(1-x^3)dx+\frac{1}{2}\int_{-1}^{1}x\,dx\int_{x^3}^{1}f(x^2+y^2)d(x^2+y^2)$$

$$=-\int_{-1}^{1}x^4\sqrt{1-x^2}\,dx+\frac{1}{2}\int_{-1}^{1}x[F(x^2+1)-F(x^2+x^6)]dx$$

$$=-2\int_{0}^{1}x^4\sqrt{1-x^2}\,dx\xlongequal{x=\sin t}-2\int_{0}^{\frac{\pi}{2}}\sin^4t\cos^2t\,dt=-2(I_4-I_6)=-\frac{\pi}{16}.$$

15. 解 $\int_{0}^{a}dx\int_{0}^{x}\frac{f'(y)}{\sqrt{(a-x)(x-y)}}dy=\int_{0}^{a}f'(y)dy\int_{y}^{a}\frac{1}{\sqrt{(a-x)(x-y)}}dx$. 而

$$\int_{y}^{a}\frac{1}{\sqrt{(a-x)(x-y)}}dx=\int_{y}^{a}\frac{d\left(x-\frac{a+y}{2}\right)}{\sqrt{\left(\frac{a-y}{2}\right)^2-\left(x-\frac{a+y}{2}\right)^2}}=\arcsin\frac{x-\frac{a+y}{2}}{\frac{a-y}{2}}\Big|_{y}^{a}=\pi,$$

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于是$\int_0^a \mathrm{d}x\int_0^x \frac{f'(y)}{\sqrt{(a-x)(x-y)}}\mathrm{d}y=\pi\int_0^a f'(y)\mathrm{d}y=\pi[f(a)-f(0)]$.

16. 证明 因为积分区域关于直线 $y=x$ 对称，所以$\iint\limits_D \frac{f(x)}{f(y)}\mathrm{d}x\,\mathrm{d}y=\iint\limits_D \frac{f(y)}{f(x)}\mathrm{d}x\,\mathrm{d}y$，于是

$$\iint\limits_D \frac{f(x)}{f(y)}\mathrm{d}x\,\mathrm{d}y=\frac{1}{2}\iint\limits_D\left[\frac{f(x)}{f(y)}+\frac{f(y)}{f(x)}\right]\mathrm{d}x\,\mathrm{d}y.$$

又因为 $f(x)>0$，所以$\frac{f(x)}{f(y)}+\frac{f(y)}{f(x)}\geqslant 2$，从而

$$\iint\limits_D \frac{f(x)}{f(y)}\mathrm{d}x\,\mathrm{d}y=\frac{1}{2}\iint\limits_D\left[\frac{f(x)}{f(y)}+\frac{f(y)}{f(x)}\right]\mathrm{d}x\,\mathrm{d}y\geqslant\iint\limits_D \mathrm{d}x\,\mathrm{d}y=(b-a)^2.$$

17. 证明 令 $F(x)=\int_a^x f(t)\mathrm{d}t$，则

$$\begin{aligned}\int_a^b f(x)\mathrm{d}x\int_x^b f(y)\mathrm{d}y&=\int_a^b f(x)[F(b)-F(x)]\mathrm{d}x\\&=F(b)\int_a^b f(x)\mathrm{d}x-\int_a^b f(x)F(x)\mathrm{d}x\\&=F^2(b)-\int_a^b F(x)\mathrm{d}F(x)\\&=F^2(b)-\frac{1}{2}F^2(x)\Big|_a^b=\frac{1}{2}F^2(b)=\frac{1}{2}\left[\int_a^b f(x)\mathrm{d}x\right]^2.\end{aligned}$$

18. 证明 因为 $f(x,y)$ 在 D 上连续，所以 $f(x,y)$ 在 D 上取到最大值 M 和最小值 m，故 $m\leqslant f(x,y)\leqslant M$，又由 $g(x,y)\geqslant 0$，得

$$mg(x,y)\leqslant f(x,y)g(x,y)\leqslant Mg(x,y),$$

积分得

$$m\iint\limits_D g(x,y)\mathrm{d}\sigma\leqslant\iint\limits_D f(x,y)g(x,y)\mathrm{d}\sigma\leqslant M\iint\limits_D g(x,y)\mathrm{d}\sigma.$$

(1) 当$\iint\limits_D g(x,y)\mathrm{d}\sigma=0$ 时，$\iint\limits_D f(x,y)g(x,y)\mathrm{d}\sigma=0$，则对任意的$(\xi,\eta)\in D$，有

$$\iint\limits_D f(x,y)g(x,y)\mathrm{d}\sigma=f(\xi,\eta)\iint\limits_D g(x,y)\mathrm{d}\sigma.$$

(2) 当$\iint\limits_D g(x,y)\mathrm{d}\sigma>0$ 时，由

$$m\iint\limits_D g(x,y)\mathrm{d}\sigma\leqslant\iint\limits_D f(x,y)g(x,y)\mathrm{d}\sigma\leqslant M\iint\limits_D g(x,y)\mathrm{d}\sigma,$$

得 $m\leqslant\dfrac{\iint\limits_D f(x,y)g(x,y)\mathrm{d}\sigma}{\iint\limits_D g(x,y)\mathrm{d}\sigma}\leqslant M.$

由介值定理，存在$(\xi,\eta)\in D$，使得 $f(\xi,\eta)=\dfrac{\iint\limits_D f(x,y)g(x,y)\mathrm{d}\sigma}{\iint\limits_D g(x,y)\mathrm{d}\sigma}$，即

$$\iint\limits_D f(x,y)g(x,y)\mathrm{d}\sigma=f(\xi,\eta)\iint\limits_D g(x,y)\mathrm{d}\sigma.$$

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19. 证明 $\dfrac{\int_0^1 xf^2(x)\mathrm{d}x}{\int_0^1 xf(x)\mathrm{d}x}\leqslant\dfrac{\int_0^1 f^2(x)\mathrm{d}x}{\int_0^1 f(x)\mathrm{d}x}$ 等价于$\int_0^1 f^2(x)\mathrm{d}x\int_0^1 xf(x)\mathrm{d}x\geqslant\int_0^1 f(x)\mathrm{d}x\int_0^1 xf^2(x)\mathrm{d}x$，

等价于$\int_0^1 f^2(x)\mathrm{d}x\int_0^1 yf(y)\mathrm{d}y\geqslant\int_0^1 f(x)\mathrm{d}x\int_0^1 yf^2(y)\mathrm{d}y$，可得

$$\int_0^1\mathrm{d}x\int_0^1 yf(x)f(y)[f(x)-f(y)]\mathrm{d}y\geqslant 0.$$

令 $I=\int_0^1\mathrm{d}x\int_0^1 yf(x)f(y)[f(x)-f(y)]\mathrm{d}y$，根据对称性，可得

$$I=\int_0^1\mathrm{d}x\int_0^1 xf(x)f(y)[f(y)-f(x)]\mathrm{d}y,$$

$$2I=\int_0^1\mathrm{d}x\int_0^1 f(x)f(y)(y-x)[f(x)-f(y)]\mathrm{d}y,$$

因为 $f(x)>0$ 且单调减少，所以$(y-x)[f(x)-f(y)]\geqslant 0$，于是 $2I\geqslant 0$，或 $I\geqslant 0$，

所以$\dfrac{\int_0^1 xf^2(x)\mathrm{d}x}{\int_0^1 xf(x)\mathrm{d}x}\leqslant\dfrac{\int_0^1 f^2(x)\mathrm{d}x}{\int_0^1 f(x)\mathrm{d}x}$.

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